



Introducing EMOPARKNZ: the Emotional Speech Database from New Zealand English Speakers with Parkinson's Disease

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Abstract

This paper introduces EMOPARKNZ, a new speech database designed to study emotional speech production in New Zealand English speakers with Parkinson's disease. The database features 13 speakers (7 male, 6 female) and 1,950 speech recordings expressing five primary emotions: excited, angry, happy, sad, and neutral. The design considerations, participant selection, and recording procedures are detailed. Acoustic analysis of the database revealed significant differences in fundamental frequency, intensity, and speech rate across emotional states. A preliminary perception test with 22 New Zealand English listeners achieved 63% classification accuracy across the five emotions, aligning with the performance reported for similar emotional speech databases. The results indicate EMOPARKNZ is a valuable resource for advancing research in emotion communication for individuals with Parkinson's disease and speech emotion classification.

Index Terms: Emotional Speech Corpus, New Zealand English, Parkinson's Disease, Acoustic Analysis, Perception Test

1. Introduction

Parkinson's disease (PD) is a progressive neurodegenerative movement disorder that significantly impacts motor function and quality of life [1]. Speech disorders are among the most prevalent symptoms, affecting an estimated 70–90% of individuals with PD over the course of the disease, with hypokinetic dysarthria being the most common [2]. Hypokinetic dysarthria impairs the phonatory, articulatory, respiratory, and prosodic components of speech.

Emotions play a central role in human communication, shaping how individuals express themselves and connect with others [3]. Inability to express emotions not only hinders social interactions but also exacerbate feelings of isolation and emotional distress, further diminishing quality of life [4]. PD significantly affects emotional speech production, where people with PD often exhibit reduced modulation of pitch and intensity, altered speech rate, and imprecise articulation, resulting in monotonous or hypophonic speech [4]. Despite the importance of understanding how PD affects emotional speech production, research on this topic remains limited, particularly for New Zealand English speakers.

Epidemiological studies indicate that age is the primary risk factor for PD, with prevalence increasing significantly in older populations [5]. In Western countries like New Zealand, increasing life expectancy has led to a rapidly ageing demographic. Projections from the 2023 New Zealand census suggest that individuals aged 65 years and older will comprise 32% of the population by 2073 [6]. As PD incidence rises alongside this demographic shift, there is an urgent need to better under-

stand the disease impact on communication [7], particularly in the context of New Zealand English.

Existing emotional speech databases primarily focus on healthy speakers or populations outside of New Zealand, overlooking the linguistic and prosodic features unique to New Zealand English. The scarcity of PD-focused corpora is understandable, as clinical studies face challenges in recruiting participants. However, the quality of emotion recognition research is heavily influenced by the design of the speech corpus, which must consider naturalness, linguistic context, and scope [8]. To address these gaps, we developed EMOPARKNZ, a novel emotional speech database specifically designed to capture the vocal characteristics of New Zealand English speakers with PD. By analysing fundamental frequency, intensity, and speech rate, this research aims to delineate the alterations in speech attributable to PD and provide insights into the interplay between physiological changes and vocal expression in affected individuals.

While the acoustic analysis provides valuable insights into the speech characteristics of individuals with PD, perceptual validation is crucial for understanding how these alterations affect listeners' ability to recognise emotions. We conducted a human perception test of EMOPARKNZ, in which listeners classified emotional expressions in speech samples from the database. This study offers essential insights into how accurately humans can identify and interpret emotional cues in Parkinsonian speech, complementing the acoustic analysis with real-world listener responses [9]. The perceptual test serves as a validation measure for the accuracy, authenticity and reliability of the dataset [10].

2. Corpus Design

2.1. Scope

A widely adopted design approach in emotion recognition research is to focus on a concise set of primary emotions, facilitating structured and comparable analyses [8]. This study examines five primary emotions: angry, happy, sad, excited, and neutral. The inclusion of a neutral or "no emotion" control condition as a baseline is common in studies of emotional speech [11]. The selection of these specific emotions was guided by current research trends in emotional speech analysis, ensuring the database is both relevant and comparable to existing resources [12].

2.2. Speakers

The corpus comprises speech recordings from thirteen individuals diagnosed with PD, seven males and six females, all native speakers of New Zealand English. Male participants ranged in

Table 1: Age, severity, and time after the PD diagnosis of male and female patients.

M-PD			F-PD		
Age	Severity	T	Age	Severity	T
56	mild-mod	6	50	mild	4
62	mild-mod	8	53	mild	7
65	mild-mod	4	55	mild	6
67	mild-mod	20	74	mild-mod	3
73	mild	4	74	mild-mod	7
74	mild	14	80	mild-mod	10
49	severe	21			

M-PD: Male-PD, F-PD: Female-PD, T: time post PD diagnosis (years)
Severity: mild/mild-moderate (mild-mod)/severe.

age from 49 to 74 years (mean 63.7 ± 9.0 years), and for female participants, 50 to 80 years (mean 64.3 ± 13.9 years). All participants were professionally diagnosed with PD by a neurologist and were prescribed medication to manage symptoms. To ensure consistency in speech sample collection, participants were recorded within three hours of taking their medication, ensuring participants were in the ON-state, when the medication was still effective [13]. This time frame is important, as the medication’s effectiveness diminishes over time. Disease severity was classified by the first author based on the overall impact of PD on daily functioning. Details regarding age, severity, and years since diagnosis for each speaker are provided in Table 1.

2.3. Stimuli

Emotions were elicited using a variation of the Stanislavski method [14], in which specific situations and corresponding images were presented to speakers to evoke and sustain the desired emotional state. The approach aligned each speaker’s mental state with the intended emotions throughout the recording session. Emotions were recorded in sequence from high to low arousal: excited, angry, happy, neutral, sad. To ensure consistency, the same leading phrases and images were provided to all speakers, who were instructed to maintain consistent emotional intensity. The sentences and prompts used in this study were adapted from those in the JL corpus [15], a related NZ English speech dataset developed for strictly-guided emotions. Fifteen emotionally neutral sentences were chosen (e.g. “Jack views an art piece”), each consisting of 4-7 syllables to minimise potential emotional drift that may occur in longer phrases. The complete list of sentences used is available in [16].

2.4. Recording procedure

Ethical approval for this study was granted by the University of Auckland Human Participants Ethics Committee (UAH-PEC23603). The recording setup involved an omnidirectional condenser microphone (AKG, HC 577L) positioned 7 cm at a 45-degree angle to the right of the speaker’s mouth. This microphone was connected to a Microsoft Surface Laptop 3 via a pre-amplifier (M-Audio MobilePre [MK II]), with audio capture at a 44.1kHz sample rate with a 16-bit quantisation. Recordings took place in a sound-treated room within the Speech Science laboratory at the University of Auckland, New Zealand. Participants were seated comfortably and faced an iPad displaying both the sentence prompts and corresponding emotional images. This arrangement ensured participants were well-acquainted with the recording environment. Before starting, participants were given five minutes to familiarise themselves with the setup and ask questions. The session began with a warm-up task, in which participants read the sentence

“Tom beats that farmer” at high intensity. This exercise allowed for the adjustment of amplification gain levels to prevent clipping, with gain settings fixed for the duration of the session. The main recording task followed, where participants read sentences displayed on the monitor, each paired with an image representing the intended emotion. The first author monitored the session in real-time to ensure technical quality. Each sentence prompt was changed upon completion of each emotion expression. Recording sessions lasted up to one hour, and the entire process was repeated to ensure two variations of each sentence for each emotion, resulting in a comprehensive corpus. In total, the study produced 1950 primary emotion sentences, encapsulating responses from 13 participants across 5 primary emotions, each repeated twice with 15 different sentences, all part of the EMOPARKNZ corpus.

2.5. Methodological challenges

Emotional speech databases are typically categorised into three types: natural, acted (simulated), and induced (elicited). Natural emotional speech samples are ideal because they capture real interactions. However, the databases usually have a limited number of speakers and suffer from reduced control, short voice samples, and poor audio quality [17]. In contrast, acted speech databases involve professional actors portraying emotions, providing greater control, repeatability, and simplicity. However, they are frequently criticised for being exaggerated or artificial [18, 19]. A third approach elicits emotions through stimuli such as images, music, or memory recall. While widely used, these methods can be passive, failing to engage participants as real-life emotions do, and may not trigger the physiological responses that influence speech [20]. Ethical concerns also restrict the use of techniques that induce high-intensity emotions [9]. Additionally, individual differences in perception and mood introduce variability, limiting consistency across recordings. These challenges complicate comparisons of vocal expressions across different database types [21]. The trade-off between spontaneity and experimental control remains a key challenge. Spontaneous emotions offer authenticity but are difficult to capture, while acted portrayals provide control at the expense of naturalness [9]. Addressing these methodological issues is essential for creating reliable speech corpora that advance emotion recognition research.

3. Emotional Corpus Acoustic Analysis

3.1. Feature extraction

To investigate the impact of PD on vocal expression, six acoustic features were analysed: fundamental frequency, intensity, speech rate, zero-crossing rate, shimmer, and jitter. This paper focuses on fundamental frequency, intensity, and speech rate, as these features demonstrated the strongest differentiation between emotions. The audio files were imported into Python (v3.12.0) for initial processing, where leading and trailing silences were removed using the pydub package [22]. Fundamental frequency values were extracted using the summation of residual harmonics method, implemented in the *covarep* package [23] in MATLAB [24]. Intensity measurements (in dB) were obtained using the intensity function in the phonetic software Praat [25]. Speech rate, measured in syllables per second, was determined by dividing the total number of syllables in a sentence by its duration, following the methodology in [26]. For each acoustic feature, mean values were calculated per sentence and aggregated per speaker for each primary emotion.

Table 2: Mean and standard deviation (SD) pitch values for EMOPARKNZ speakers ($n=12$).

Emotion		Excited	Angry	Happy	Neutral	Sad
Total Pitch (Hz)	Mean	247.83	216.11	219.83	173.72	182.79
	SD	65.24	64.01	61.14	39.52	50.77
Total Intensity (dB)	Mean	60.65	61.25	57.13	52.40	51.01
	SD	2.75	3.65	2.70	2.57	3.22
Total Speech Rate	Mean	2.86	2.63	2.86	2.85	2.76
	SD	0.31	0.33	0.25	0.21	0.36

Table 3: Statistical analysis comparison of emotion pairs.

Emotion Pairs	Fundamental Frequency		Intensity		Speech Rate	
	t-ratio	p-value	t-ratio	p-value	t-ratio	p-value
angry - excited	-15.413	<.0001	3.242	0.0106	-12.473	<.0001
angry - happy	-1.808	0.3692	22.139	<.0001	-12.830	<.0001
angry - neutral	20.604	<.0001	47.601	<.0001	-12.011	<.0001
angry - sad	16.195	<.0001	55.078	<.0001	-7.034	<.0001
excited - happy	13.605	<.0001	18.897	<.0001	-0.357	0.9965
excited - neutral	36.017	<.0001	44.360	<.0001	0.462	0.9906
excited - sad	31.607	<.0001	51.836	<.0001	5.439	<.0001
happy - neutral	22.412	<.0001	25.462	<.0001	0.819	0.9248
happy - sad	18.002	<.0001	32.939	<.0001	5.796	<.0001
neutral - sad	-4.410	0.0001	7.477	<.0001	4.977	<.0001

Statistical analysis was conducted using a linear mixed-effects model implemented in the R environment [27] with the *lme4* package [28]. Emotion and gender were set as fixed effects, while speaker identity and sentence were random effects. Consult [16] for a more detailed description of the model.

3.2. Results

The mean fundamental frequency, intensity, and speech rate for each emotion, averaged across all speakers, are shown in Table 2. Table 3 presents statistical comparisons between emotion pairs across three acoustic parameters. Statistical analysis using a linear mixed-effects model revealed significant interactions for most emotion pairs with respect to fundamental frequency, except between angry and happy. The model also identified a significant two-way interaction between emotion and gender ($\chi^2_{(4)} = 1195$, $p < 0.0001$), where females displayed higher F_0 values across all emotions. For intensity, the model found significant interactions for all emotion pairs and a significant two-way interaction between emotion and gender ($\chi^2_{(4)} = 1195$, $p < 0.0001$). Males exhibited higher intensity values than females for all emotions except angry. The linear mixed-effects model also showed that speech rate is significantly affected by emotion ($\chi^2_{(4)} = 226$, $p < 0.001$). Speech for angry was significantly slower than for all other emotions ($p < 0.0001$). While gender did not significantly affect speech rate, a trend suggested that males tended to speak faster than females across all emotions.

4. Emotional Corpus Evaluation

4.1. Listeners

A human emotion perception test was completed by 22 listeners (8 male, 14 female), all New Zealand English speakers aged 18 years or older. Among the male listeners, 1 was aged 20-29,

6 were aged 30-39, and 1 was aged 40-49. Among the female listeners, 5 were aged 20-29, 4 were aged 30-39, 3 were aged 40-49, and 2 were aged 50-59. Only 2 female listeners reported having experience with voice changes in individuals with PD. None of the listeners reported any hearing deficits.

4.2. Listening Tasks

The preliminary human perception test was conducted on a subset of the data. Three speakers were selected to represent each PD severity level. The perception test consisted of 97 questions and took approximately 15 to 25 minutes to complete. It began with an introductory slide outlining the purpose of the test and informed participants that they would be listening to speech samples from individuals diagnosed with PD. The first seven questions gathered demographic information, including age and gender. Participants were also asked about any history of hearing loss and prior experience interacting with individuals with PD. The remaining 90 questions were divided into three sets of 30 questions, each featuring audio samples from a different speaker with varying degrees of disease progression (mild, mild-moderate, and severe). Prior to the emotion classification task, participants listened to a neutral sentence spoken by each speaker to familiarise themselves with the speaker's vocal style. This step was not a training phase, as participants were not asked to classify the neutral sentence into emotion categories but simply listened to a sentence categorised as neutral. The inclusion of a neutral sentence before the perception test, as per Tóth et al. [9], was found not to bias emotion recognition accuracy in listeners. In total, 90 stimulus sentences (6 sentences $\times$ 5 emotions $\times$ 3 speakers; specifically, sentences 1, 5, 6, 9, 11, and 15 as described in [16]) were presented to the listeners. The test was completed using a computer or mobile device with headphones. The speech stimuli were randomised to mitigate potential effects of presentation order. The process was self-paced, allowing listeners to replay the recordings as many times as necessary.

4.3. Results

The results presented in Table 4 summarise the human perception test of emotions involving three distinct speakers. Each column represents the classification accuracy for one of these speakers, while the overall column reflects the aggregated results across all three speakers, providing a broader view of emotion recognition performance across varying PD severity levels. Negative emotions were generally well recognised across all PD severity levels, with sad being the most accurately classified emotion, achieving an overall accuracy of 82%. Angry was also recognised well, with an overall accuracy of 71%. In contrast, positive emotions such as happy and excited were more challenging for listeners to identify, with accuracy of 50% each. The neutral emotion was classified with moderate accuracy at 60%, suggesting that listeners found neutral expressions moderately distinguishable across the speakers. The aggregated confusion matrix for all three speakers in Figure 1 has an overall classification accuracy of 63%. The matrix illustrates the distribution of predicted emotion labels against the true labels. Each cell represents the number of instances classified into a given category, with darker shades indicating higher counts. Frequently confused emotion pairs include sad-neutral, angry-excited, happy-excited, and happy-neutral.

Table 4: Classification accuracy for human emotion perception test.

	Mild PD	Mild-Moderate PD	Severe PD	Overall
Angry	52%	81%	81%	71%
Excited	49%	27%	74%	50%
Happy	31%	59%	59%	50%
Sad	76%	80%	90%	82%
Neutral	49%	66%	64%	60%



Figure 1: Aggregated confusion matrix of emotion classification results. Each cell represents the number of votes cast for a predicted emotion, with rows indicating the actual emotions and columns indicating the predicted emotions.

5. Discussion

The acoustic analysis demonstrates that fundamental frequency, intensity, and speech rate can be utilised to differentiate emotional states in speech produced by individuals with PD. As shown in Table 3, significant differences in fundamental frequency were found between most emotion pairs, particularly between high-arousal emotions (angry, excited, happy) and lower-arousal ones (sad). These results align with previous research, which suggests that heightened emotional arousal leads to an increase in fundamental frequency. For instance, the comparison between angry and neutral speech reveals a substantial increase in pitch for anger, supporting the idea that anger raises pitch due to increased vocal tension [29, 30]. In contrast, sad speech exhibited a lower pitch, consistent with the reduced vibratory capacity of the vocal folds observed in PD speakers [31]. Regarding intensity, our analysis shows that high-arousal emotions tend to be associated with higher mean intensity levels. This trend supports the notion that intensity, like fundamental frequency, serves as an arousal-differentiating feature, reflecting the greater energy demands these emotions place on speakers. For speech rate, the post-hoc comparisons confirmed that angry speech was significantly slower than all other emotions ($p < 0.0001$), suggesting that anger has the most pronounced impact on speech rate. While sad speech was also slower than happy, excited, and neutral speech, the difference was less pronounced. No significant differences were found between happy, excited, and neutral speech rates, implying that these emotions do not notably affect speech rate relative to each other.

The results of the perception survey provide strong preliminary evidence validating the EMOPARKNZ corpus. Listeners achieved an overall emotion recognition accuracy of 63% across five primary emotions, a result consistent with the 69% accuracy reported for the JL corpus with 60 listeners [15] and similar to recognition rates around 60% reported in prior studies [9, 32, 33]. These findings highlight that, despite the variability introduced by speakers with differing degrees of PD, listeners were able to decode emotions with relatively high accuracy, supporting the EMOPARKNZ corpus as a reliable resource for future research on speech emotion perception in PD populations. The confusion matrix in Figure 1 revealed notable variations in recognition accuracy across specific emotions. Sad and angry emotions were among the most accurately identified, aligning with previous findings [9], whereas happy emotions posed the greatest challenge. Common misclassifications included sad as neutral, angry as excited or neutral, and happy as excited or neutral. Excited emotions were frequently mistaken for happy, reflecting the inherent difficulty in distinguishing emotions with overlapping arousal characteristics. There was minimal confusion between low-arousal emotions (sad, neutral) and high-arousal emotions (excited). This finding underscores the importance of arousal in emotion perception, as suggested by the Valence-Arousal model, which posits that emotions differing in arousal are easier to distinguish than those differing primarily in valence [34]. These results indicate that arousal may play a more critical role than valence in listeners' emotion recognition, particularly when considering the effects of PD-related changes in speech. A notable methodological consideration is the gender bias in the speaker sample, as this perception test exclusively featured male speakers. This decision was made to represent all three levels of PD severity, with the only available severe PD speaker being male. Consequently, the results are limited by gender bias. However, this choice is partially justified by the higher incidence of PD among males, as men are twice as likely to develop the disease [35]. Thus, the male-only sample aligns with the demographics of the PD population.

6. Conclusion

This paper introduced the EMOPARKNZ corpus, a unique database of induced emotion speech featuring dynamic emotional expressions from speakers diagnosed with PD. With 1950 high-quality speech recordings, it offers a valuable resource for further exploration of emotional expression in PD populations. Acoustic analysis revealed significant variability in the prosodic features fundamental frequency, intensity, and speech rate across different emotions, confirming that these parameters effectively capture emotional content despite the vocal impairments associated with PD. The results from the human perception test involving untrained listeners demonstrated the validity of the database, providing strong preliminary evidence of its usefulness in speech emotion recognition research. Future research will focus on analysing time- and frequency-domain acoustic features to train machine learning models in emotion classification tasks. Additionally, the collected speech data will be used in a large-scale perceptual study involving a broader pool of listeners and incorporating all sentences from the database. Upon the completion of the first author's doctoral thesis, the EMOPARKNZ corpus will be made publicly available. This lays the foundation for future studies aimed at improving emotion recognition technologies and communication aids for people with PD.

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